

Coding & Solution

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and the overall solution. The submission demonstrates working functionality, clean code practices, and alignment with the defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Repository Link / URL	https://github.com/rajeswari-boggavarapu/A-Comprehensive-Measure-of-Well-Being
Programming Language(s)	Python 3.x, HTML5, CSS3
Framework(s) Used	Flask (backend routing & templating), scikit-learn (ML), pandas (data), Jinja2 (frontend rendering), Gunicorn (production server)
Key Features Implemented	• Country dropdown (53 countries from CSV) • Client-side form validation (required, type=number) • Dataset loading with FileNotFoundError guard • RandomForestRegressor model (300 estimators) • HDI score prediction via model.predict() • Tier classification using UNDP thresholds • Conditional result display via Jinja2 • Live deployment on Render with Gunicorn
Pending / Incomplete Features	• Auto-fill of indicator values when a country is selected • Server-side numeric range validation • Saved model persistence using joblib • Interactive data visualizations (Plotly charts)
Setup / Run Instructions	1. Clone the repository from GitHub 2. Navigate into the hdi_app folder 3. Run: pip install flask scikit-learn numpy pandas gunicorn 4. Run: python app.py 5. Open http://127.0.0.1:5000 in your browser

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes

S.No	Criteria	Status (Yes / No)
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The HDI Predictor is a fully functional end-to-end machine learning web application. The codebase is organized into clearly named functions (load_dataset, train_model, classify_hdi) following single-responsibility principles. Section comments are used throughout app.py. Error handling covers missing dataset files (FileNotFoundError) and invalid form inputs (TypeError, ValueError). The application is deployed live on Render and the full source code is available on GitHub. All 8 planned features have been implemented and tested successfully.